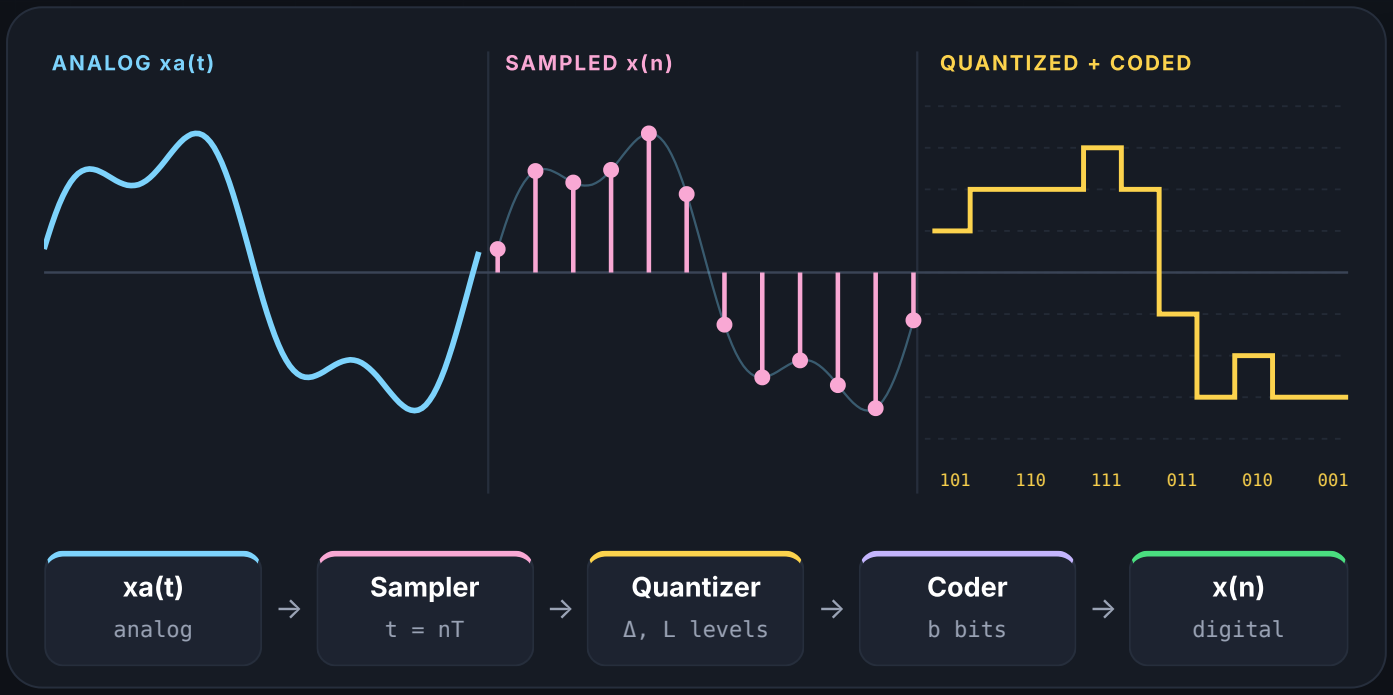
CHAPTERS

01·02 Repeated Problems in the Exams

Repeated Problems · Ch 1 & 2

Every problem that came in the previous midterms, finals and quizzes — solved step by step the doctor's way, most repeated first



10	4	14	25
Chapter 1 problems	Chapter 2 problems	all repeated 2+ times	previous exams & quizzes

CHAPTER 1
Sampling & periodicity
Lectures 2 · 3 · 4

CHAPTER 2
DFT & FFT
Lectures 5 · 6

ORDER
Most repeated first
exam count on every card

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DSP / ch01 / contents

Contents

Repeated problems · Ch 1 & Ch 2 · 14 questions

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Problems from the previous exams (sampling, aliasing and periodicity), most repeated first
10 questions p. 3

02 **Chapter 2**
Problems from the previous exams (DFT, IDFT, FFT and IFFT), most repeated first
4 questions p. 24

How to study from this file

Chapter 1 then Chapter 2, each ordered by the number of exams the problem came in. Every problem has its full solution, result box and the exams it came in.

SECTION 01

Chapter 1

Problems from the previous exams (sampling, aliasing and periodicity), most repeated first

Questions

- ☐ P7 Periodicity of continuous-time and discrete-time signals p. 3
- ☐ P8 x(t) = cos(200πt), Fs = 400 Hz p. 6
- ☐ P13 x(t) = 3 sin(100πt), Fs = 300 samples/s p. 8
- ☐ P1 Example 1.1 — Sampling & aliasing p. 10
- ☐ P10 x(t) = 10 cos(1000πt), Fs = 2000 sample/s p. 12
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P7 ● Frequent 9×

Periodicity of continuous-time and discrete-time signals

Check the periodicity of the following signals and find their fundamental periods:
(i) $x_a(t)$ = 3 cos(5t + π/6) (ii) x(n) = cos(3πn) (iii) x(n) = cos(0.01πn) (iv) X(n) = 2 sin(3πn/105) (v) x(t) = cos(3000πt) (vi) y(t) = 7 cos(400πt) (vii) x(n) = sin(2n + π/4)

(i)

xa(t) = 3 cos(5t + π/6) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{T_p = 1/f = 2\pi/5 \text{ sec}}$$

$$x_a(t + T_p) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

∴ Periodic, fundamental period Tp = 2π/5 sec.

(ii)

x(n) = cos(3πn) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi \rightarrow F = 3/2 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 2) = \cos(3\pi n + 6\pi) = \cos(3\pi n) = x(n)$$

∴ Periodic, fundamental period **N = 2 samples**.

(iii)

x(n) = cos(0.01πn) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 0.01\pi \rightarrow F = 1/200 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 200) = \cos(0.01\pi n + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 200 samples**.

(iv)

x(n) = 2 sin(3πn/105) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi/105 = \pi/35 \rightarrow F = 1/70 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 70) = 2\sin(\pi n/35 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 70 samples**.

(v)

x(t) = cos(3000πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 3000\pi \rightarrow f = 1500 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 1/1500 \text{ sec} = 0.667 \text{ msec}}$$
$$x(t + Tp) = \cos(3000\pi t + 2\pi) = x(t)$$

∴ Periodic, fundamental period Tp = 1/1500 sec = 0.667 msec.

(vi)

y(t) = 7 cos(400πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 400\pi \rightarrow f = 200 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 5 \text{ msec}}$$
$$y(t + Tp) = 7\cos(400\pi t + 2\pi) = y(t)$$

∴ Periodic, fundamental period Tp = 5 msec.

(vii)

x(n) = sin(2n + π/4) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 2 \rightarrow F = 2/2\pi = 1/\pi$$

F cannot be expressed as a ratio of two integers → **not periodic**.

Result Tp = 2π/5 s · N = 2 · N = 200 · N = 70 · Tp = 1/1500 s · Tp = 5 ms · sin(2n + π/4) not periodic

Key idea الـ continuous دايمًا periodic و Tp = 1/f. الـ discrete لازم F تبقى rational، ولو F طلع فيها π تبقى الإشارة مش periodic.

Source — from your PDFs

Lecture	**LEC 3** · Sec. 1.5.1 · p. 3
Lecture	**LEC 3** · Sec. 1.5.2 · p. 9
Came in	Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm 1 Model-2 · Midterm 8-12-2020 · Final 27-6-2021 · Final 12-1-2019 · Final 23-12-2017 · Final 2-1-2017 · Midterm OLD

P8 ● Frequent 8×

x(t) = cos(200πt), Fs = 400 Hz

For the signal x(t) = cos(200πt) to be sampled with Fs = 400 Hz:

(i) Find an expression for its corresponding sequence x(n). Then calculate its fundamental period N_P.

(ii) Find the sample values of the first cycle in x(n), then plot x(n) and x(t) on the same figure.

$$x(t) = 1\cos(200\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 1,\quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100\ Hz}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200\ Hz}$$
$$Fs = 400\ Hz \geq 2f_{max} = 200\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(i)

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

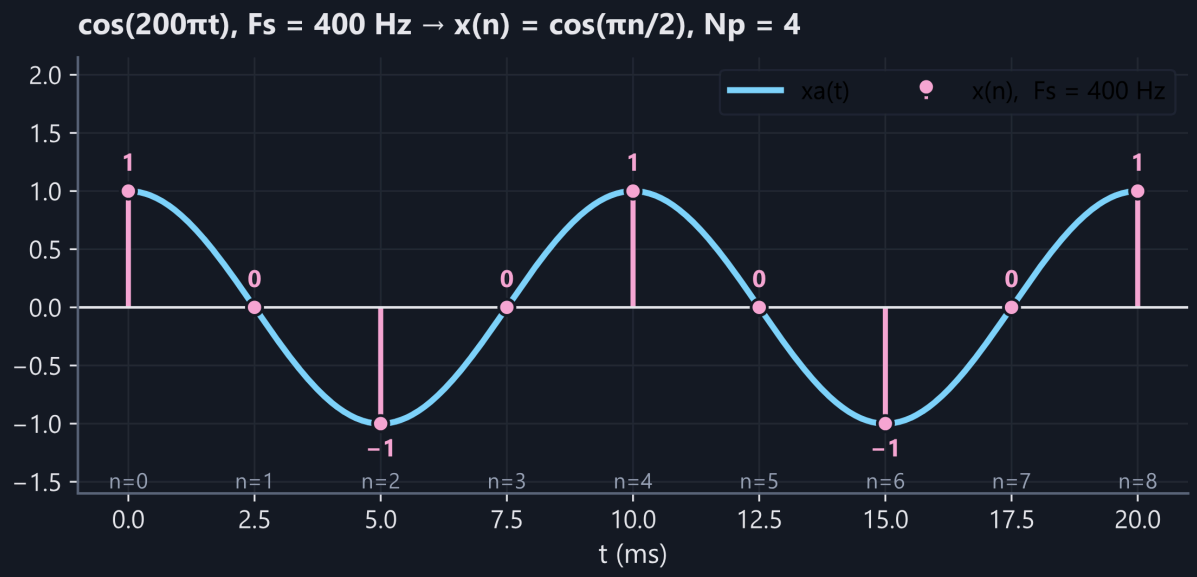
$$x(n) = x_a(nT) = x_a(n/Fs) = 1\cos(200\pi\, n/400) = \mathbf{1\cos(\pi/2\, n)}$$

$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 1\cos(0) = \mathbf{1}$$
$$x(1) = 1\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 1\cos(\pi) = \mathbf{-1}$$
$$x(3) = 1\cos(3\pi/2) = \mathbf{0}$$

(ii) Plot x(t) and x(n) on the same figure:



Result `fN = 200 Hz, no aliasing · x(n) = cos(πn/2), Np = 4: 1, 0, −1, 0`

Key idea نوع M1: f = 100 Hz و Fs = 400 أكبر من 2f، فمفيش aliasing. نعوّض t = n/Fs فنطلع F = 1/4 يعني Np = 4، ونحسب أربع samples.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm 1 Model-2 · Final 7-6-2023 · Q4-b · Final 8-9-2021 · Final 18-1-2024 · Quiz 1 · Section · Midterm OLD · several models

P13 ● Frequent 5×

x(t) = 3 sin(100πt), Fs = 300 samples/s

Consider x(t) = 3 sin(100πt):

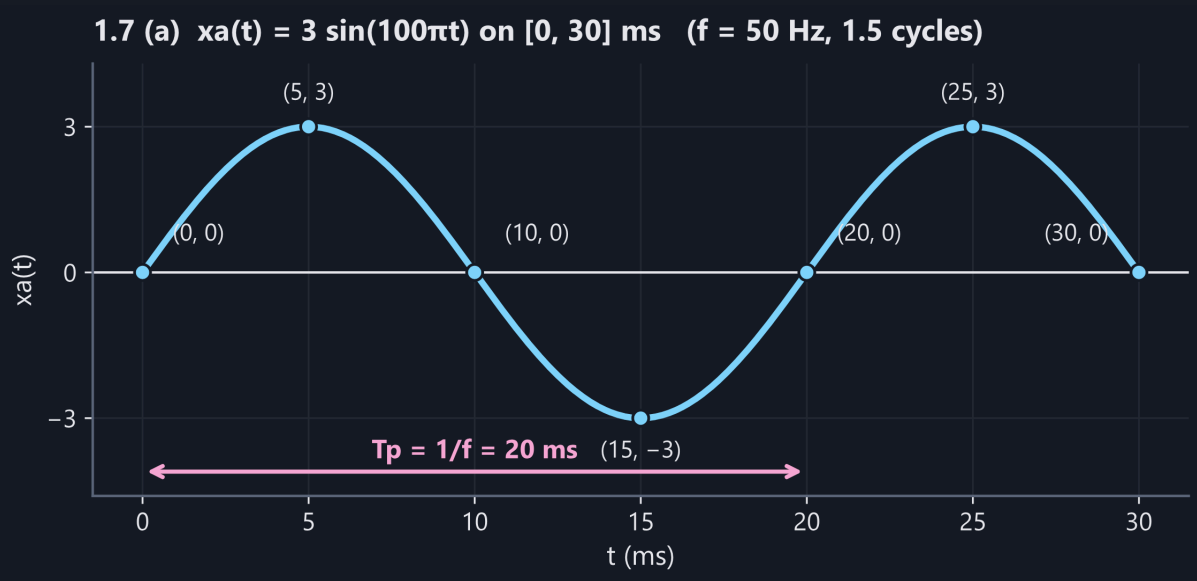
(i) Sketch the signal for [0, 30] ms.

(ii) The signal is sampled at Fs = 300 samples/s. Determine the frequency of x(n) and show that it is periodic.

(iii) Compute the sample values in one period of x(n). Sketch x(n) on the same diagram with x(t). What is the period of x(n)?

(iv) Can you find Fs such that x(n) reaches its peak value 3? What is the minimum Fs?

(i) xa(t) = 3 sin(100πt): A = 3, 2πf = 100π → f = 50 Hz, Tp = 1/f = 20 msec. The interval [0, 30] ms contains 1.5 periods.



(ii), (iii)

If the signal is sampled at Fs = 300 Hz (t = nT = n/Fs), the discrete-time signal is:

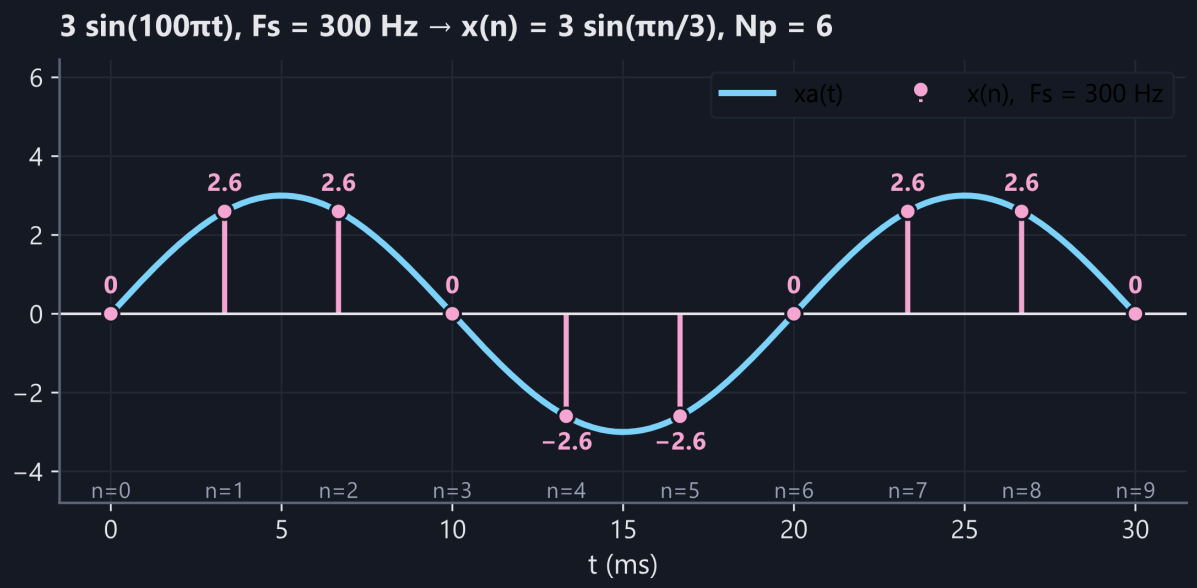
$$x(n) = x_a(nT) = x_a(n/Fs) = 3\sin(100\pi\, n/300) = \mathbf{3\sin(\pi/3\, n)}$$

$$F = f/Fs = 50/300 = \mathbf{1/6} = k/N_P \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 6 \text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 5):

$$x(0) = 3\sin(0) = \mathbf{0}$$
$$x(1) = 3\sin(\pi/3) = \mathbf{2.598}$$
$$x(2) = 3\sin(2\pi/3) = \mathbf{2.598}$$
$$x(3) = 3\sin(\pi) = \mathbf{0}$$
$$x(4) = 3\sin(4\pi/3) = \mathbf{-2.598}$$
$$x(5) = 3\sin(5\pi/3) = \mathbf{-2.598}$$

Period of x(n): N = 6 samples = 6 × (1/300) s = 20 msec.



(iv) x(n) = 3 sin(100πn/Fs) reaches its peak value 3 when:

$$\sin(100\pi n/Fs) = 1 \rightarrow 100\pi n/Fs = \pi/2 \rightarrow Fs = 200n \rightarrow \text{minimum at } n = 1: \mathbf{Fs = 200 \text{ samples/sec}}$$

Result x(n) = 3 sin(πn/3), N = 6 samples = 20 ms · peak value reached when Fs = 200 samples/s

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Midterm 28-11-2021 · Q1-b · Midterm 2 · Q1-c · Midterm 8-12-2020 · Q1-c · Final 4-3-2021 · Q1-b · Sheet 1 · 1.7

P1 3×

Example 1.1 — Sampling & aliasing

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$.

a) Determine the minimum sampling rate required to avoid aliasing.

b) The signal is sampled at Fs = 200 Hz. What is the discrete-time signal?

c) The signal is sampled at Fs = 75 Hz. What is the discrete-time signal?

d) What is the frequency $0 < f < Fs/2$ of a sinusoid that yields samples identical to those in (c)?

a) The frequency of the analog signal is f = 50 Hz. Hence the minimum sampling rate required to avoid aliasing is Fs = 100 Hz.

b) If the signal is sampled at Fs = 200 Hz, the discrete-time signal is

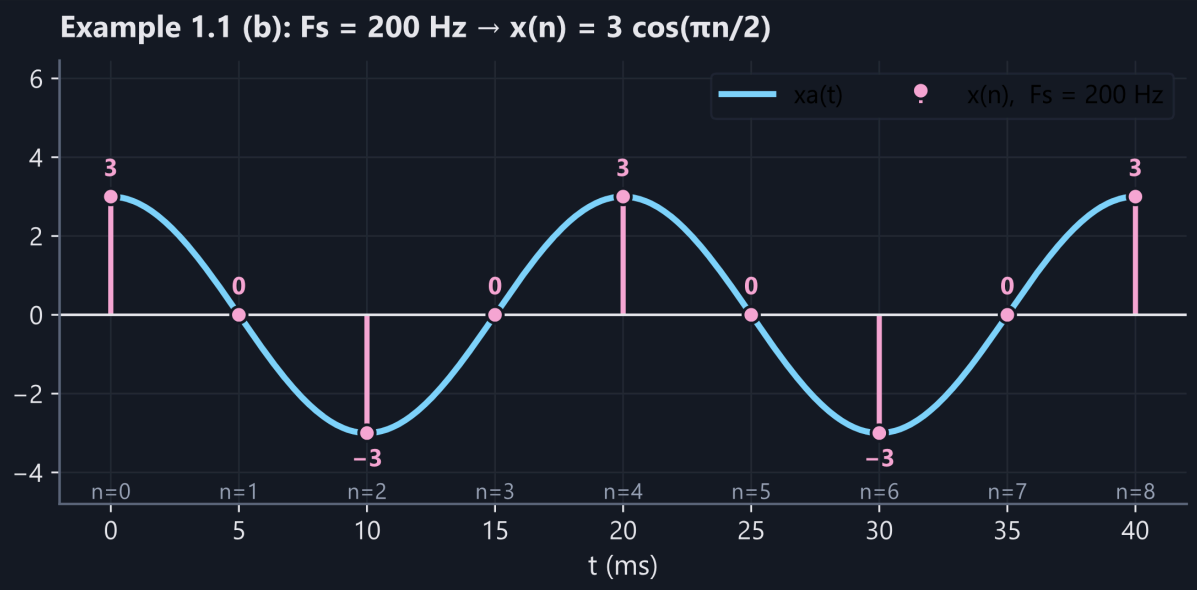
$x(n) = 3 \cos(100\pi/200)n = 3 \cos(\pi/2)n$

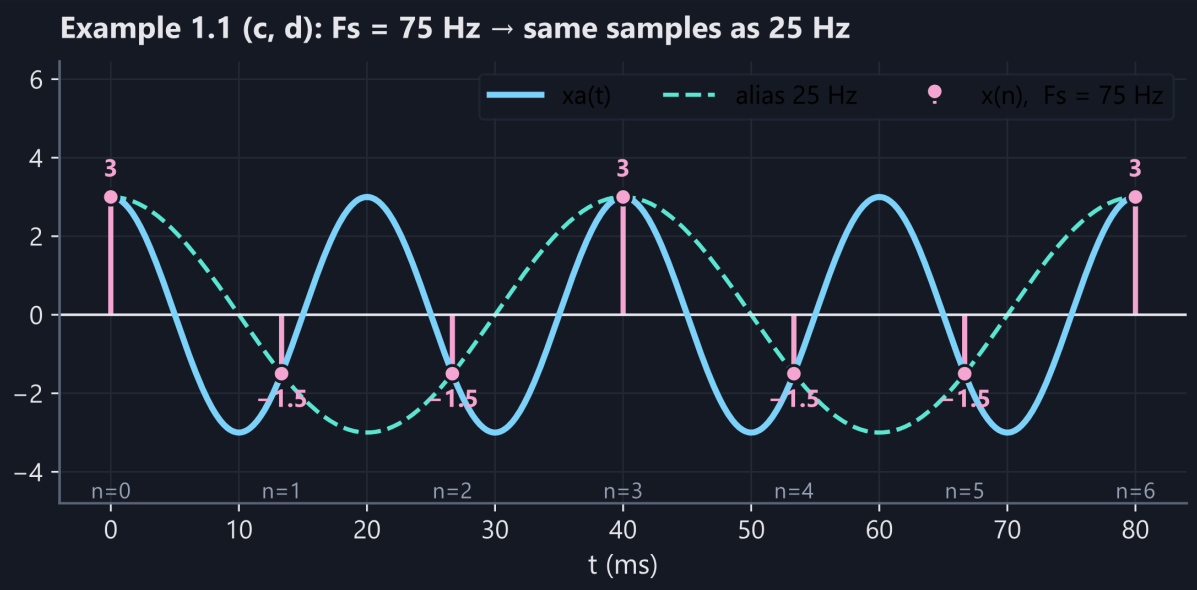
c) If the signal is sampled at Fs = 75 Hz, the discrete-time signal is

$x(n) = 3 \cos(100\pi/75)n = 3 \cos(4\pi/3)n = 3 \cos(2\pi - 2\pi/3)n = 3 \cos(2\pi/3)n$

d) For the sampling rate of Fs = 75 Hz, we have f = F·Fs = 75F. The frequency of the sinusoid in part (c) is F = 1/3. Hence f = 25 Hz.

Clearly, the sinusoidal signal $ya(t) = 3 \cos 2\pi ft = 3 \cos 50\pi t$ sampled at Fs = 75 samples/s yields identical samples. Hence f = 50 Hz is an alias of f = 25 Hz for the sampling rate Fs = 75 Hz.





Result Fs,min = 100 Hz · Fs = 200 Hz → x(n) = 3 cos(πn/2) · Fs = 75 Hz → x(n) = 3 cos(2πn/3) · 25 Hz gives the same samples (50 Hz is its alias)

Key idea نطلع f من المعادلة ونقارن Fs بـ 2f. لو Fs أقل يحصل aliasing، والتردد اللي بيظهر هو اللي بيدّي نفس الـ samples جوه المدى من 0 لحد Fs/2.

Source — from your PDFs

Lecture **LEC 4** · Example 1.1 · p. 38

Came in Final 20-1-2025 | Final 11-6-2023 | Final 23-1-2023 (same type: is Fs enough? + x(n))

P10 3×

x(t) = 10 cos(1000πt), Fs = 2000 sample/s

For x(t) = 10 cos(1000πt):

(i) If sampled with Fs = 2000 sample/s, is it possible to reconstruct it successfully?

(ii) Find the Nyquist frequency F_N.

(iii) Using F_N, find an expression for x(n).

(iv) Find the values of x(n) in the first cycle.

$$x(t) = 10\cos(1000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1000\pi \rightarrow \mathbf{f = 500\ Hz}$$

(i), (ii)

$$F_N = 2f_{max} = 2 \times 500 = \mathbf{1000\ Hz}$$
$$Fs = 2000\ Hz \geq 2f_{max} = 1000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(iii) Using the obtained Nyquist frequency F_N = 1000 Hz:

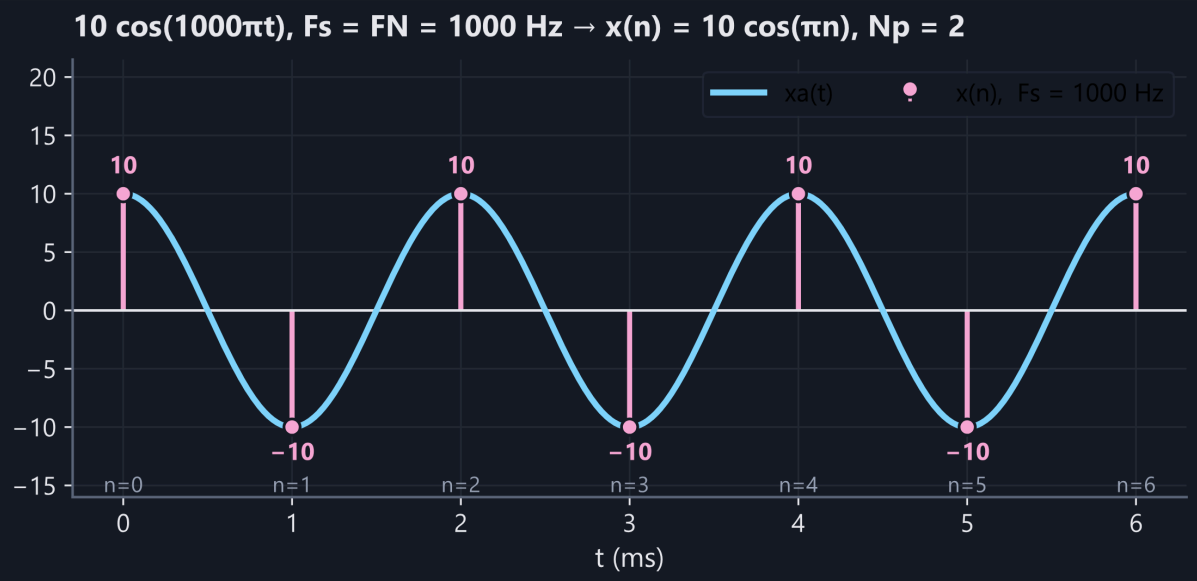
If the signal is sampled at F_N = 1000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 10\cos(1000\pi\, n/1000) = \mathbf{10\cos(\pi\, n)}$$

$$F = f/Fs = 500/1000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi) = \mathbf{-10}$$



Result fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10

Key idea نوع M1 بس المطلوب نستخدم الـ Nyquist frequency: FN = 2f = 1000 Hz، فـ F = 1/2 و Np = 2، والـ samples بتطلع ±10 على القمم بس.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 11-6-2023 · Q1-b · Final 20-1-2025 · Q3-c · Final 23-1-2023 · Q4-b

P11 ● Frequent 4×

$x(t) = A\cos(2\pi f_o t)$, $A = 2$ V, $f_o = 10$ kHz

For $x_a(t) = A\cos(2\pi f_o t)$, $A = 2$ volt, $f_o = 10$ kHz:

a) Check the periodicity and find the fundamental period.

b) If sampled at Fs = 5 kHz, is it possible to reconstruct the original signal?

c) What is the minimum sampling frequency?

d) Using it, find x(n) and the sample values during one cycle (or the first four samples).

$$x_a(t) = A\cos(2\pi f_0 t),\ A = 2\text{ V},\ f_0 = 10\text{ kHz} \rightarrow x_a(t) = 2\cos(20000\pi t)$$

a) For every fixed value of f, xa(t) is periodic: **$T_p = 1/f_o = 1/10^4 = 0.1$ msec**.

b)

$$F_N = 2f_{max} = 2 \times 10 = \mathbf{20\text{ kHz}}$$
$$Fs = 5\text{ kHz} < 2f_{max} = 20\text{ kHz}$$

Since $Fs < 2f_{max}$, aliasing occurs and the signal **cannot be reconstructed** from its samples.

c) The minimum sampling frequency is **$Fs = 2f_{max} = 20$ kHz**.

d)

If the signal is sampled at Fs = 20000 Hz (t = nT = n/Fs), the discrete-time signal is:

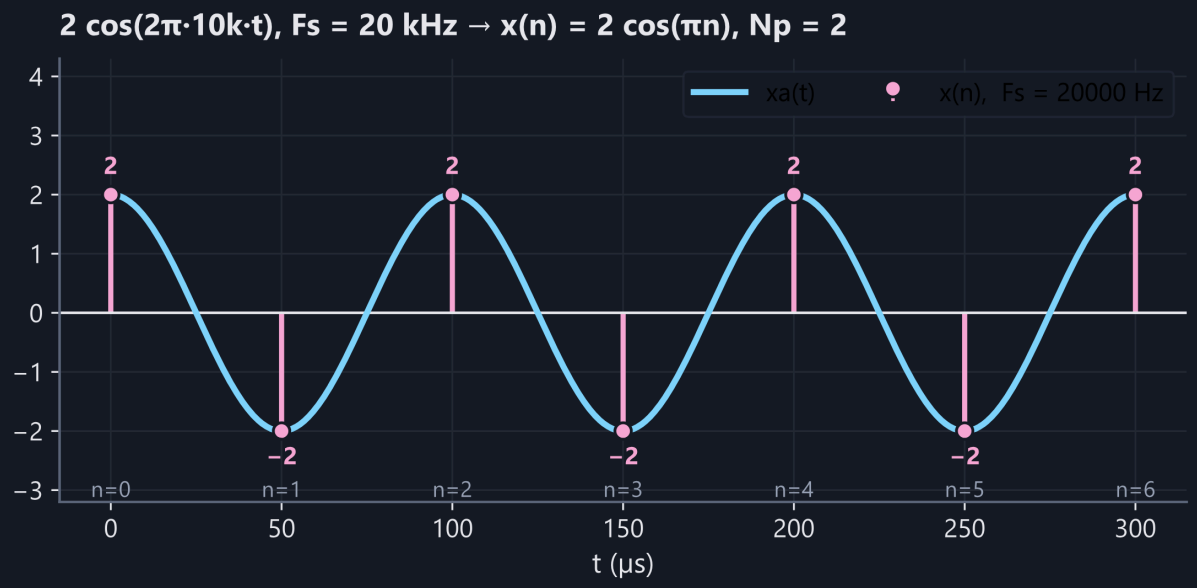
$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(20000\pi\, n/20000) = \mathbf{2\cos(\pi\, n)}$$

$$F = f/Fs = 10000/20000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$

First four samples: x(0) = 2, x(1) = −2, x(2) = 2, x(3) = −2



Result Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2

Key idea Fs = 5 kHz أقل من 2f = 20 kHz فيحصل aliasing. أقل Fs يرجّع الإشارة هو 20 kHz، وعنده كل الـ samples بتطلع ±2.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 9-6-2026 · Q1-b · Final 17-8-2026 · Q1-b · Quiz 1 Summer 26 · Q2 · Revision Sheet 1 · Q10

P2 3×

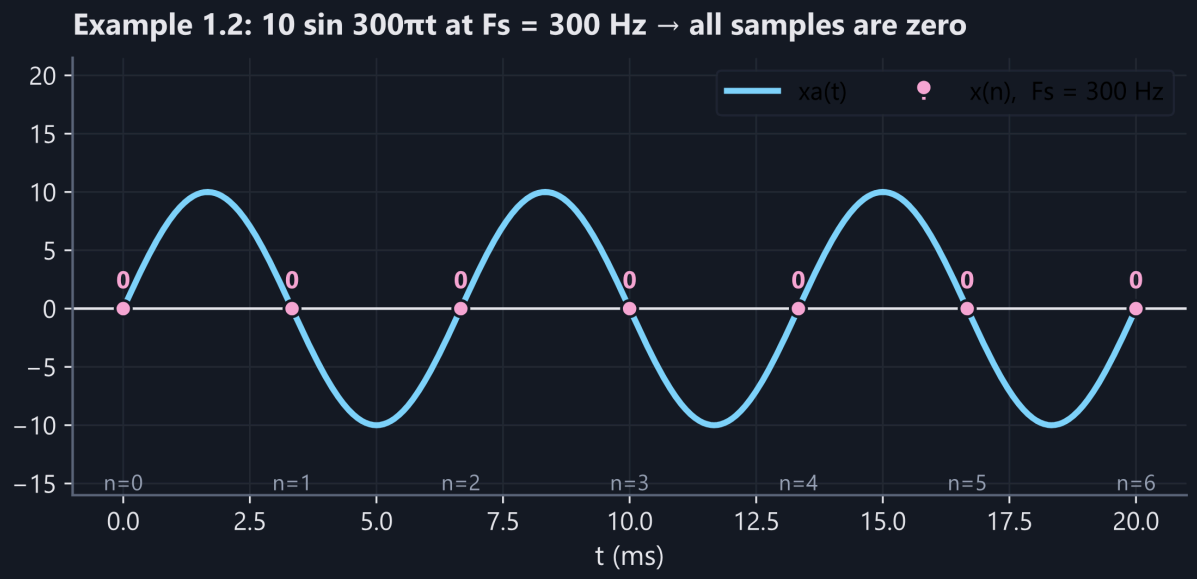
Example 1.2 — Nyquist rate

Consider $x_a(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$. What is the Nyquist rate for this signal?

The frequencies present in the signal above are f1 = 25 Hz, f2 = 150 Hz, f3 = 50 Hz. Thus fmax = 150 Hz and according to (1.47), Fs > 2fmax = 300 Hz. The Nyquist rate is fN = 2fmax. Hence fN = 300 Hz.

Discussion

- It should be observed that the signal component 10 sin 300πt, sampled at the Nyquist rate fN = 300, results in the samples 10 sin πn, which are identically zero.
- In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely.
- This situation would not occur if the sinusoid is offset in phase by some amount θ. In such a case we have 10 sin(300πt + θ) sampled at the Nyquist rate fN = 300 samples per second, which yields the samples:
  10 sin(πn + θ) = 10(sin πn cos θ + cos πn sin θ) = 10 sin θ cos πn = $(-1)^n$ 10 sin θ
- Thus if θ ≠ 0 or π, the samples of the sinusoid taken at the Nyquist rate are not all zero.
- However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown.
- A simple remedy that avoids this potentially troublesome situation is to sample the analog signal at a rate higher than the Nyquist rate.



Result fN = 300 Hz; sampling exactly at fN can miss the 150 Hz component (samples = $(-1)^n$ 10 sin θ) → sample above the Nyquist rate

Key idea نطلع كل الترددات اللي في الإشارة ونختار أكبرها fmax، والـ Nyquist rate يساوي 2fmax.

Source — from your PDFs

Lecture **LEC 4** · Example 1.2 · p. 41

Came in Final 12-1-2019 | Final 28-12-2019

P9 3×

x(t) = 10 cos(1500πt), Fs = 3000 samples/s

For x(t) = 10 cos(1500πt):
(i) If it is sampled with Fs = 3000 sample/s, is it possible to reconstruct this signal from its samples? Find the Nyquist frequency F_N.
(ii) Using Fs, find x(n), then calculate the sample values during one cycle and plot x(t) and x(n).

$$x(t) = 10\cos(1500\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1500\pi \rightarrow \mathbf{f = 750\ Hz}$$

(i)

$$F_N = 2f_{max} = 2 \times 750 = \mathbf{1500\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 1500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(ii)

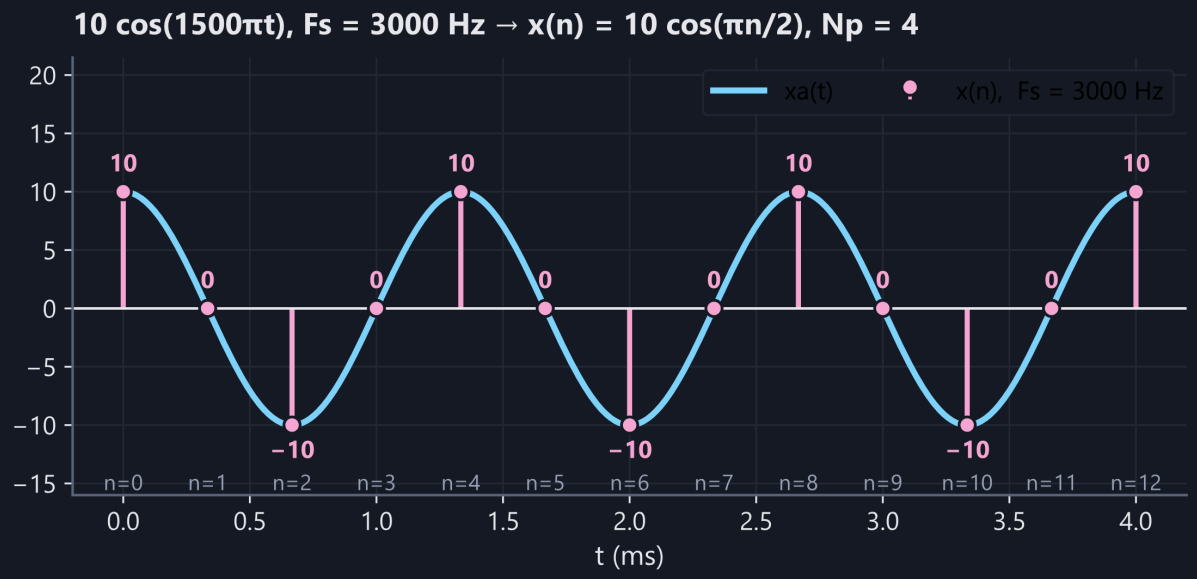
If the signal is sampled at Fs = 3000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 10\cos(1500\pi\, n/3000) = \mathbf{10\cos(\pi/2\, n)}$$

$$F = f/Fs = 750/3000 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 10\cos(\pi) = \mathbf{-10}$$
$$x(3) = 10\cos(3\pi/2) = \mathbf{0}$$



Result `fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0`

Key idea M1: f = 750 Hz نوع و Fs = 3000 أكبر من 2f، فالإشارة ترجع. F = 1/4 يعني Np = 4، والـ samples هي 10 و0 و−10 و0.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 12-1-2019 · Final 28-12-2019 · Revision Sheet 1 · Q9

P12 2×

x(t) = 2 cos(2000πt), Fs = 3000 Hz then Fs = 8000 Hz

Given x(t) = 2 cos(2000πt) to be sampled with Fs = 3000 Hz:

(i) Could the original signal be reconstructed successfully from its samples?

(ii) Take Fs = 8000 Hz; find x(n) and its fundamental period N_P.

(iii) Find the sample values of the first cycle and plot x(n).

$$x(t) = 2\cos(2000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 2,\quad 2\pi f = 2000\pi \rightarrow \mathbf{f = 1000\ Hz}$$

(i)

$$F_N = 2f_{max} = 2 \times 1000 = \mathbf{2000\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 2000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(ii), (iii)

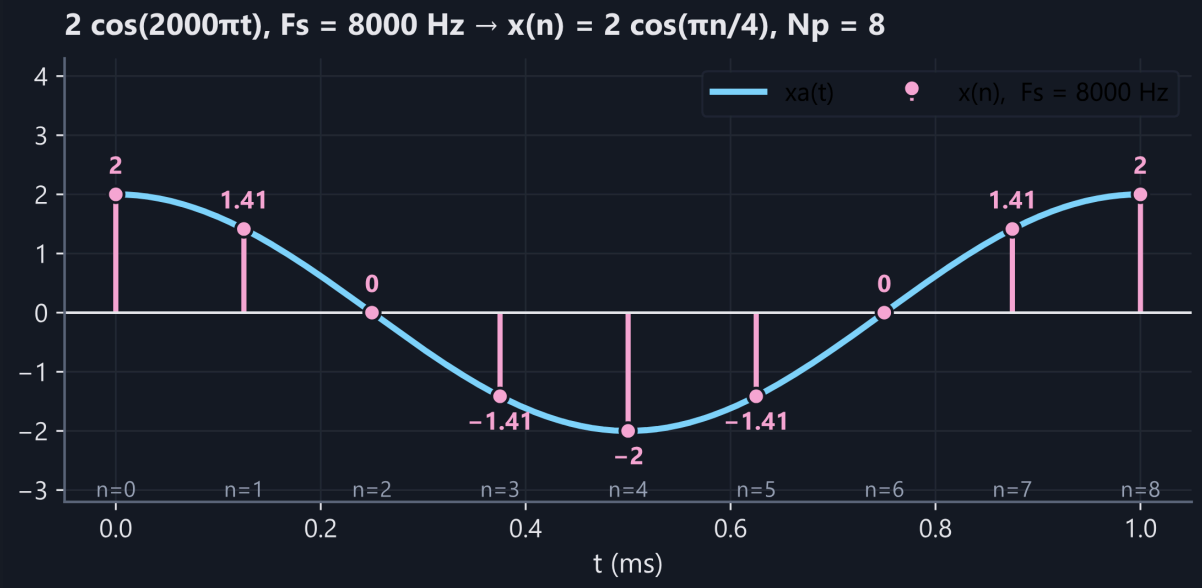
If the signal is sampled at Fs = 8000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(2000\pi\, n/8000) = \mathbf{2\cos(\pi/4\, n)}$$

$$F = f/Fs = 1000/8000 = \mathbf{1/8} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 8\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 7):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi/4) = \mathbf{1.414}$$
$$x(2) = 2\cos(\pi/2) = \mathbf{0}$$
$$x(3) = 2\cos(3\pi/4) = \mathbf{-1.414}$$
$$x(4) = 2\cos(\pi) = \mathbf{-2}$$
$$x(5) = 2\cos(5\pi/4) = \mathbf{-1.414}$$
$$x(6) = 2\cos(3\pi/2) = \mathbf{0}$$
$$x(7) = 2\cos(7\pi/4) = \mathbf{1.414}$$



Result No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414

Key idea مع Fs = 3000 الإشارة ترجع لأنها أكبر من 2f = 2000. ومع Fs = 8000 بيبقى F = 1/8 يعني Np = 8 samples في الـ cycle.

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Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Midterm 12-11-2025 · Q1 · Midterm Solve 14-11-2019 · Q1-c

P14 3×

Is Fs suitable? (Fs = Nyquist rate exactly)

(a) x(t) = 2 cos(3000πt), Fs = 3 kHz. (b) x(t) = 2 cos(2000πt), Fs = 2 kHz. (c) x(t) = cos(300πt), Fs = 300 Hz.

Is Fs suitable? If so, find x(n), the values in the first cycle, and plot.

(a)

$$x(t) = 2\cos(3000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 2,\quad 2\pi f = 3000\pi \rightarrow \mathbf{f = 1500\ Hz}$$

$$F_N = 2f_{max} = 2 \times 1500 = \mathbf{3000\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 3000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

$Fs = 2f_{max}$ exactly (the Nyquist rate): it is the minimum rate. Sampling at a rate higher than the Nyquist rate is preferred (Example 1.2).

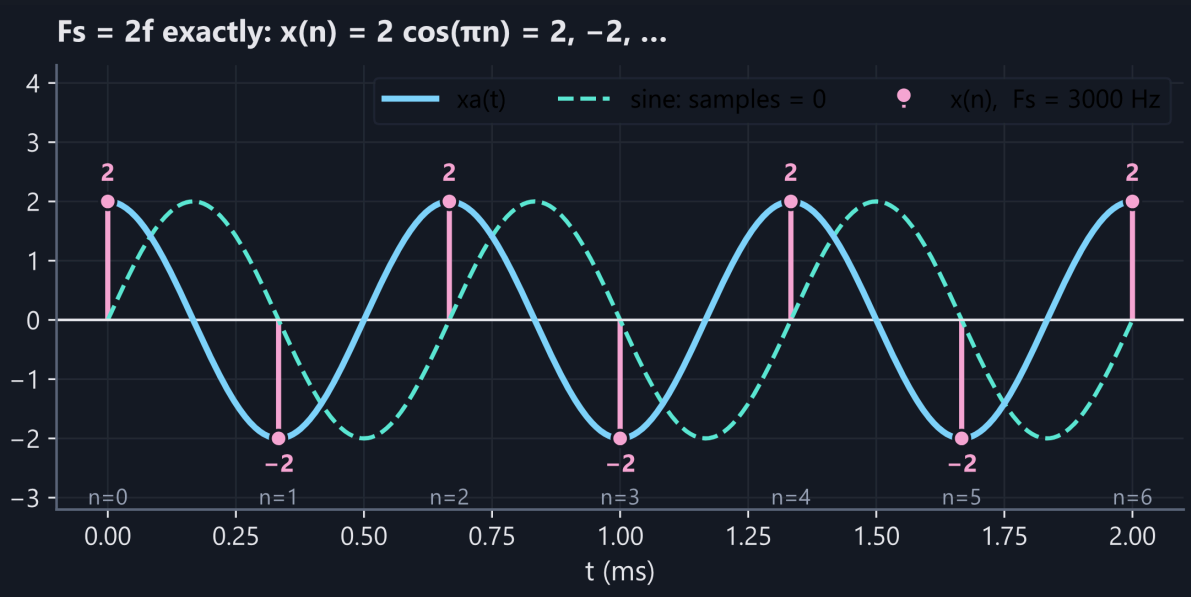
If the signal is sampled at Fs = 3000 Hz ($t = nT = n/Fs$), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(3000\pi\, n/3000) = \mathbf{2\cos(\pi\, n)}$$

$$F = f/Fs = 1500/3000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$



(b) $x(t) = 2\cos(2000\pi t)$, Fs = 2 kHz → f = 1000 Hz, Fs = 2f → $x(n) = 2\cos(\pi n)$: 2, −2.

(c) $x(t) = \cos(300\pi t)$, Fs = 300 Hz → f = 150 Hz, Fs = 2f → $x(n) = \cos(\pi n)$: 1, −1.

Result Fs = 2fmax exactly → x(n) = 2 cos(πn): 2, −2 (works, but a rate above Nyquist is preferred)

Key idea لما Fs = 2f بالظبط الـ samples بتقع على القمم بس (±A). ده ينفع نظريًا، بس الأحسن نعمل sampling أسرع من الـ Nyquist rate.

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Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 2-1-2017 · Q1-c · Final 23-12-2017 · Q1-c · Book Midterm 2015 Model-A

★ Cheat sheet · Chapter 1

Final results

P7 Tp = 2π/5 s · N = 2 · N = 200 · N = 70 · Tp = 1/1500 s · Tp = 5 ms · sin(2n + π/4) not periodic

P8 fN = 200 Hz, no aliasing · x(n) = cos(πn/2), Np = 4: 1, 0, −1, 0

P13 x(n) = 3 sin(πn/3), N = 6 samples = 20 ms · peak value reached when Fs = 200 samples/s

P1 Fs,min = 100 Hz · Fs = 200 Hz → x(n) = 3 cos(πn/2) · Fs = 75 Hz → x(n) = 3 cos(2πn/3) · 25 Hz gives the same samples (50 Hz is its alias)

P10 fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10

P11 Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2

P2 fN = 300 Hz; sampling exactly at fN can miss the 150 Hz component (samples = $(-1)^n$ 10 sin θ) → sample above the Nyquist rate

P9 fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0

P12 No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414

P14 Fs = 2fmax exactly → x(n) = 2 cos(πn): 2, −2 (works, but a rate above Nyquist is preferred)

SECTION 02

Chapter 2

Problems from the previous exams (DFT, IDFT, FFT and IFFT), most repeated first

Questions

- ☐ **P9** x(n) = {1, 0, 1, 1} p. 24
- ☐ **P10** x(n) = {1, 0, 1, 0} p. 29
- ☐ **P12** x(n) = {1, 0, 0, 1} p. 34
- ☐ **P11** x(n) = {1, 1, 0, 1} p. 39

P9 ● Frequent 10×

x(n) = {1, 0, 1, 1}

For the sequence x(n) = {1, 0, 1, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 1 + 0 + 1 + 1$$
$$= 3$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + 0 + e^{-j\pi} + e^{-j3\pi/2}$$
$$= 1 + [\cos(\pi) - j\,\sin(\pi)] + [\cos(3\pi/2) - j\,\sin(3\pi/2)]$$
$$= 1 + (-1) + j$$
$$= \mathbf{j}$$

Thus X(1) = j, of magnitude 1 and phase angle Φ(1) = 90°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + 0 + e^{-j2\pi} + e^{-j3\pi}$$
$$= 1 + [\cos(2\pi) - j\,\sin(2\pi)] + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + 1 + (-1)$$
$$= \mathbf{1}$$

Thus X(2) = 1, of magnitude 1 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi) - j\,\sin(3\pi)] + [\cos(9\pi/2) - j\,\sin(9\pi/2)]$$
$$= 1 + (-1) + (-j)$$
$$= \mathbf{-j}$$

Thus X(3) = −j, of magnitude 1 and phase angle Φ(3) = −90°.

So, the time series x(n) = {1, 0, 1, 1} has the DFT given by **X(k) = {3, j, 1, −j}**.

FFT algorithm (as Example 2.3):

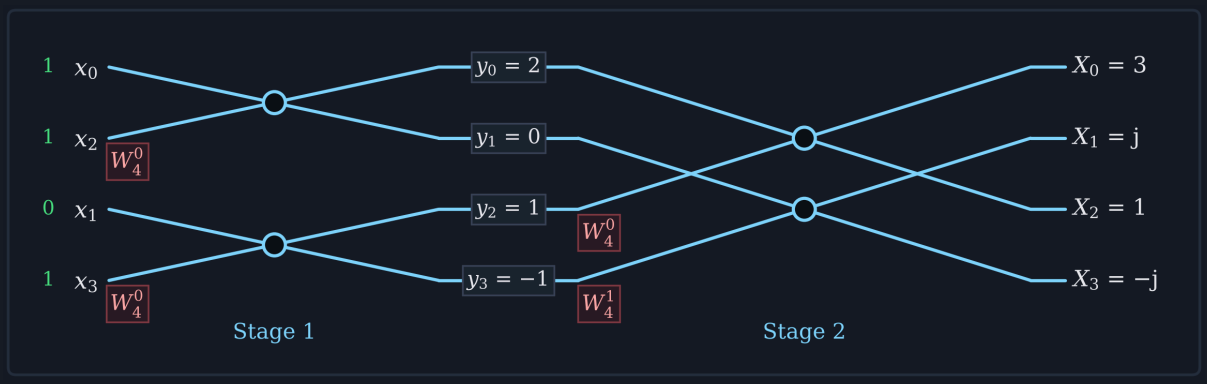
This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 1, and

$$X1(k) = X11(k) + W_N^{k}\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\ X22(0) = 2 + 1 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\ X22(1) = 0 + e^{-j\pi/2}(-1) = 0 - j(-1) = \mathbf{j}$$
$$X11(2) = X21(0) - W_8^0\ X22(0) = 2 - 1 = \mathbf{1}$$
$$X11(3) = X21(1) - W_8^2\ X22(1) = 0 - e^{-j\pi/2}(-1) = 0 + j(-1) = \mathbf{-j}$$



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + jj2] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - jj2] = \mathbf{1}$$

$\therefore$ x(n) = **{1, 0, 1, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j\pi/2} + 1\ e^{j\pi} + (-j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(j) + 1(-1) + (-j)(-j)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j\pi} + 1\ e^{j2\pi} + (-j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + j(-1) + 1(1) + (-j)(-1)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j3\pi/2} + 1\ e^{j3\pi} + (-j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(-j) + 1(-1) + (-j)(j)]$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

$\therefore$ x(n) = **{1, 0, 1, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{3, 1, 1, 1\}$

$$\text{L.H.S.} = 1^2 + 0^2 + 1^2 + 1^2 = 3$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = 3$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	j	1	−j
$\|X(k)\|$	3	1	1	1
Φ(k)	0°	90°	0°	−90°

Spectra of {1, 0, 1, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Result $X(k) = \{3, j, 1, -j\}$ · IDFT → {1, 0, 1, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.

Key idea أكتر مسألة متكررة في الشابتر: DFT أو FFT، وبعدها check بالـ IDFT أو IFFT، وبعدها Parseval، والـ spectra، والـ saving. نفس الخطوات في كل امتحان.

Key words (iii) Parseval's theorem

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 20-1-2025 · Q4-c; Final 23-1-2023 · Q3-b; Final 17-8-2026 · Q2-b; Final 18-1-2024 · Q2-c; Final 12-1-2019 · Q2-c; Final 2-1-2017 · Q2-c; Midterm 31-7-2023 · Q2-b; Midterm 8-12-2020 · Q2-c; Midterm 2 · Q2-c; Book Assignment A2.2

P10 · Frequent · 4×

x(n) = {1, 0, 1, 0}

For the sequence x(n) = {1, 0, 1, 0}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 1 + 0 + 1 + 0$$
$$= 2$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + 0 + e^{-j\pi} + 0$$
$$= 1 + [\cos(\pi) - j\,\sin(\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(1) = 0, of magnitude 0 and phase angle Φ(1) = 0°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + 0 + e^{-j2\pi} + 0$$
$$= 1 + [\cos(2\pi) - j\,\sin(2\pi)]$$
$$= 1 + 1$$
$$= \mathbf{2}$$

Thus X(2) = 2, of magnitude 2 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + 0$$
$$= 1 + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(3) = 0, of magnitude 0 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 0, 1, 0} has the DFT given by **X(k) = {2, 0, 2, 0}**.

FFT algorithm (as Example 2.3):

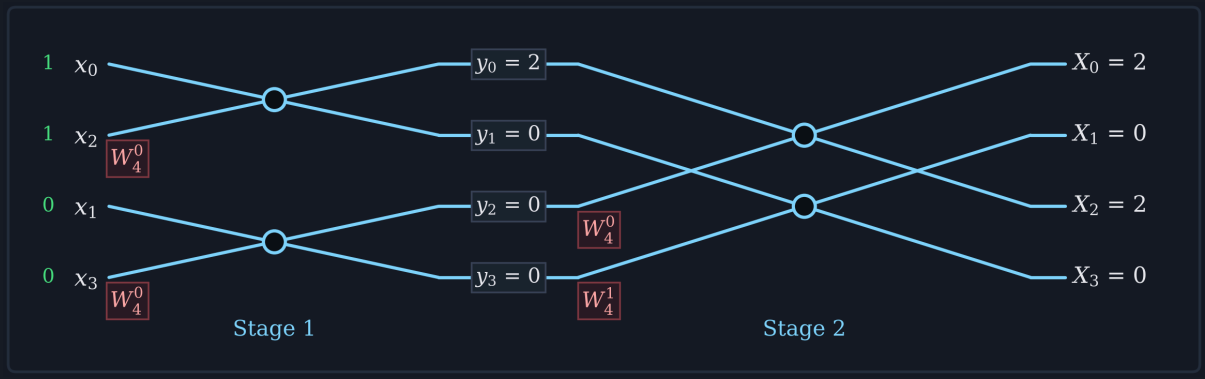
This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 0, and

$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 0 = 0$$
$$X22(1) = x1 - x3 = 0 - 0 = 0$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 2 + 0 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 0 + e^{-j\pi/2}(0) = 0 - j(0) = \mathbf{0}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 2 - 0 = \mathbf{2}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 0 - e^{-j\pi/2}(0) = 0 + j(0) = \mathbf{0}$$



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 0$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[0 + j0] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[0 - j0] = \mathbf{0}$$

$\therefore$ x(n) = **{1, 0, 1, 0}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT} = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$\begin{aligned} x(nT) = x(0) &= (1/N) \sum_{k=0}^{3} X(k)\, e^{j2\pi k(0)/4} \\ &= \tfrac{1}{4}\,[2 + 0 + 2 + 0] \\ &= \mathbf{1}, \text{ as expected.} \end{aligned}$$

With n = 1:

$$\begin{aligned} x(nT) = x(T) &= (1/N) \sum_{k=0}^{3} X(k)\, e^{j2\pi k(1)/4} \\ &= \tfrac{1}{4}\,[2 + 0 + 2\, e^{j\pi} + 0] \\ &= \tfrac{1}{4}\,[2 + 2(-1)] \\ &= \tfrac{1}{4}\,[2 + (-2)] \\ &= \mathbf{0}, \text{ as expected.} \end{aligned}$$

With n = 2:

$$\begin{aligned} x(nT) = x(2T) &= (1/N) \sum_{k=0}^{3} X(k)\, e^{j2\pi k(2)/4} \\ &= \tfrac{1}{4}\,[2 + 0 + 2\, e^{j2\pi} + 0] \\ &= \tfrac{1}{4}\,[2 + 2(1)] \\ &= \tfrac{1}{4}\,[2 + 2] \\ &= \mathbf{1}, \text{ as expected.} \end{aligned}$$

With n = 3:

$$\begin{aligned} x(nT) = x(3T) &= (1/N) \sum_{k=0}^{3} X(k)\, e^{j2\pi k(3)/4} \\ &= \tfrac{1}{4}\,[2 + 0 + 2\, e^{j3\pi} + 0] \\ &= \tfrac{1}{4}\,[2 + 2(-1)] \\ &= \tfrac{1}{4}\,[2 + (-2)] \\ &= \mathbf{0}, \text{ as expected.} \end{aligned}$$

∴ x(n) = **{1, 0, 1, 0}**, the original sequence.

(iii) Parseval's theorem:

$$\sum_{n=0}^{N-1} x^2(n) = (1/N) \sum_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, 0, 2, 0\}$

$$\text{L.H.S.} = 1^2 + 0^2 + 1^2 + 0^2 = 2$$
$$\text{R.H.S.} = (1/4)[2^2 + 0^2 + 2^2 + 0^2] = (1/4)(8) = 2$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	0	2	0
$\|X(k)\|$	2	0	2	0
Φ(k)	0°	0°	0°	0°

Spectra of {1, 0, 1, 0}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Result X(k) = {2, 0, 2, 0} · IDFT → {1, 0, 1, 0} · Parseval 2 = 2 · saving 12 mult., 4 add.

Key idea نفس الخطوات. الـ X(k) هنا كلها أرقام حقيقية {2, 0, 2, 0}، لأن الـ sequence متكررة كل 2 samples.

Key words (iii) Parseval's theorem

Source — from your PDFs

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Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 11-6-2023 · Q1-c; Final 7-6-2023 · Q1-c; Midterm 12-11-2025 · Q2-b; Midterm 28-11-2021 · Q2-b

P12 3×

x(n) = {1, 0, 0, 1}

For the sequence x(n) = {1, 0, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 1 + 0 + 0 + 1$$
$$= \mathbf{2}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + 0 + 0 + e^{-j3\pi/2}$$
$$= 1 + [\cos(3\pi/2) - j\,\sin(3\pi/2)]$$
$$= 1 + j$$
$$= \mathbf{1 + j}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + 0 + 0 + e^{-j3\pi}$$
$$= 1 + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(9\pi/2) - j\,\sin(9\pi/2)]$$
$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

FFT algorithm (as Example 2.3):

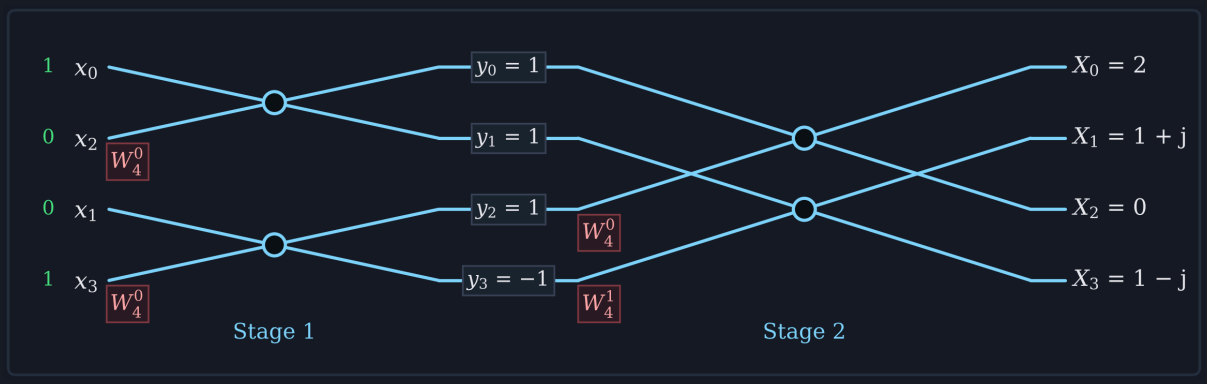
This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\ X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\ X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\ X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\ X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 2 \qquad x22(1) = X(1) - X(3) = j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + jj2] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - jj2] = \mathbf{1}$$

$\therefore$ x(n) = **{1, 0, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j) + 0 + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi/2} + 0 + (1 - j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= \tfrac{1}{4}\ [2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi} + 0 + (1 - j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j3\pi/2} + 0 + (1 - j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-j) + (1 - j)(j)]$$
$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$

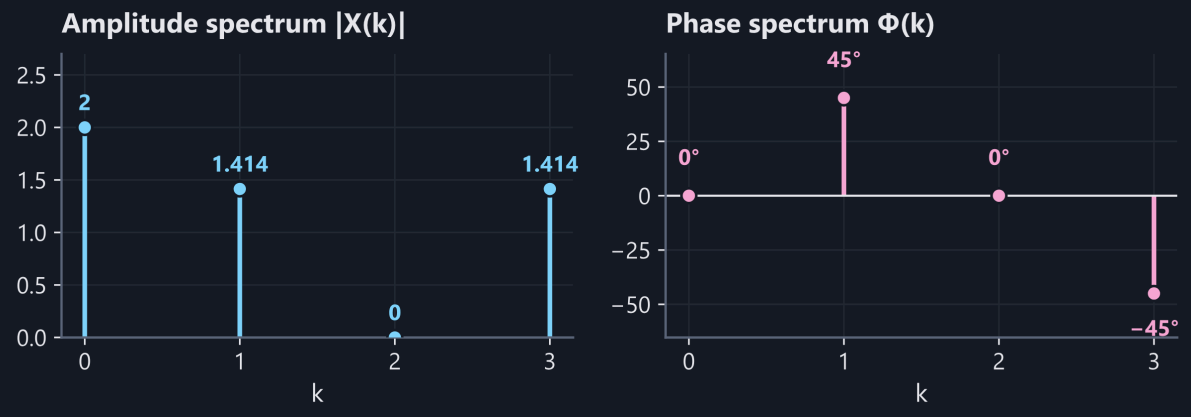
$$\text{L.H.S.} = 1^2 + 0^2 + 0^2 + 1^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
$\|X(k)\|$	2	√2 = 1.414	0	√2 = 1.414
Φ(k)	0°	45°	0°	−45°

Spectra of {1, 0, 0, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Result $X(k) = \{2, 1 + j, 0, 1 - j\}$ · IDFT → $\{1, 0, 0, 1\}$ · Parseval 2 = 2 · saving 12 mult., 4 add.

Key idea دي نفس sequence مثال 2.1، فال $X(k) = \{2, 1 + j, 0, 1 - j\}$، وبنكمّل بال IFFT والـ Parseval والـ spectra.

Key words (iii) Parseval's theorem

Source — from your PDFs

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Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 23-12-2017 · Q2-c · Final 28-12-2019 · Q2-b · Midterm 26-7-2026 · Q2-b

P11 2×

x(n) = {1, 1, 0, 1}

For the sequence x(n) = {1, 1, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 1, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 1 + 0 + 1 \\ &= 3 \end{aligned}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + e^{-j\pi/2} + 0 + e^{-j3\pi/2}$$
$$= 1 + [\cos(\pi/2) - j\sin(\pi/2)] + [\cos(3\pi/2) - j\sin(3\pi/2)]$$
$$= 1 + (-j) + j$$
$$= \mathbf{1}$$

Thus X(1) = 1, of magnitude 1 and phase angle Φ(1) = 0°.

With k = 2:

$$X(2) = \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + e^{-j\pi} + 0 + e^{-j3\pi}$$
$$= 1 + [\cos(\pi) - j\sin(\pi)] + [\cos(3\pi) - j\sin(3\pi)]$$
$$= 1 + (-1) + (-1)$$
$$= \mathbf{-1}$$

Thus X(2) = −1, of magnitude 1 and phase angle Φ(2) = 180°.

With k = 3:

$$X(3) = \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + e^{-j3\pi/2} + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi/2) - j\sin(3\pi/2)] + [\cos(9\pi/2) - j\sin(9\pi/2)]$$
$$= 1 + j + (-j)$$
$$= \mathbf{1}$$

Thus X(3) = 1, of magnitude 1 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 1, 0, 1} has the DFT given by **X(k) = {3, 1, −1, 1}**.

FFT algorithm (as Example 2.3):

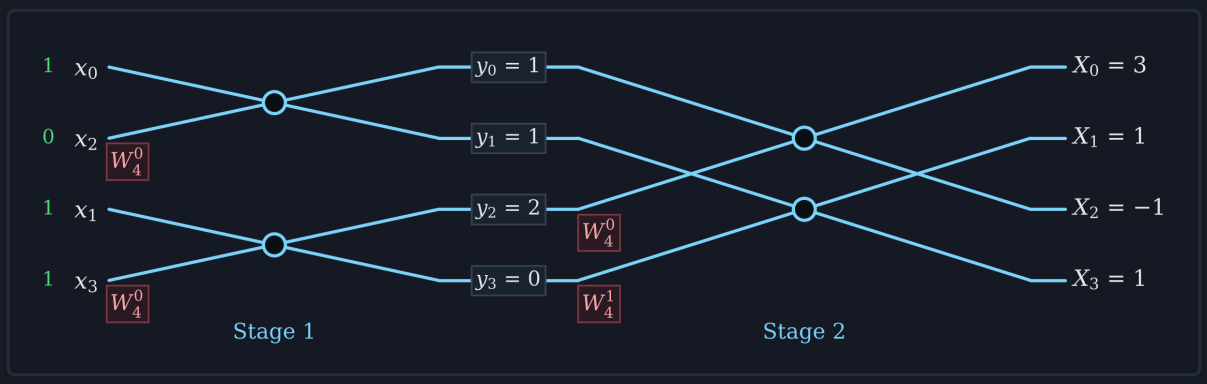
This is a four-point DFT with x0 = 1, x1 = 1, x2 = 0, x3 = 1, and

$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 1 + 1 = 2$$
$$X22(1) = x1 - x3 = 1 - 1 = 0$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 2 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(0) = 1 - j(0) = \mathbf{1}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 2 = \mathbf{-1}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(0) = 1 + j(0) = \mathbf{1}$$



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 4$$
$$x22(0) = X(1) + X(3) = 2 \qquad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[4 + j0] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[4 - j0] = \mathbf{1}$$

$\therefore$ x(n) = **{1, 1, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j\pi/2} + (-1)\ e^{j\pi} + 1\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(j) + (-1)(-1) + 1(-j)]$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j\pi} + (-1)\ e^{j2\pi} + 1\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + 1(-1) + (-1)(1) + 1(-1)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j3\pi/2} + (-1)\ e^{j3\pi} + 1\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(-j) + (-1)(-1) + 1(j)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 1, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{3, 1, 1, 1\}$

$$\text{L.H.S.} = 1^2 + 1^2 + 0^2 + 1^2 = 3$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = 3$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	1	−1	1
$\|X(k)\|$	3	1	1	1
Φ(k)	0°	0°	180°	0°

Spectra of {1, 1, 0, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Result $X(k) = \{3, 1, -1, 1\}$ · IDFT → $\{1, 1, 0, 1\}$ · Parseval 3 = 3 · saving 12 mult., 4 add.

Chapter 2

Key idea نفس الخطوات. الـ X(k) = {3, 1, −1, 1} حقيقية، لأن الـ sequence متماثلة حوالين n = 0.

Key words (iii) Parseval's theorem

Source — from your PDFs

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Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 9-6-2026 · Q2-b · Midterm 1 Model-2 · Q2-c

★ Cheat sheet · Chapter 2

Final results

- **P9** fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0
- **P10** fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10
- **P12** No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414
- **P11** Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2